

**13.10** Solution: (a) Using Eq. (13.25),

$$\Phi(\mathbf{k}, \omega) = \frac{2ze}{\epsilon(\omega)} \frac{\delta(\omega - \mathbf{k} \cdot \mathbf{v})}{k^2 - \frac{\omega^2}{c^2} \epsilon(\omega)},$$

and taking the Fourier transform, we have

$$\begin{aligned} \Phi(\omega, \mathbf{x}) &= \frac{1}{(2\pi)^{3/2}} \int d^3k e^{i\mathbf{k} \cdot \mathbf{x}} \Phi(\mathbf{k}, \omega) \\ &= \frac{1}{(2\pi)^{3/2}} \frac{2ze}{\epsilon(\omega)} \int dk_x \int dk_y \int dk_z e^{i(k_x x + k_y y + k_z z)} \frac{\delta(\omega - k_z v)}{k^2 - \frac{\omega^2}{c^2} \epsilon(\omega)} \\ &= \frac{ze}{v\epsilon(\omega)} \frac{1}{2\pi^{3/2}} \int dk_x \int dk_y e^{i\omega z/v} \frac{e^{i(k_x x + k_y y)}}{k_x^2 + k_y^2 + \frac{\omega^2}{v^2} - \frac{\omega^2}{c^2} \epsilon(\omega)} \\ &= \frac{ze}{v\epsilon(\omega)} \frac{1}{2\pi^{3/2}} \int_0^{2\pi} d\phi \int_0^{+\infty} k_{\perp} dk_{\perp} e^{i\omega z/v} \frac{e^{ik_{\perp} \rho \cos \phi}}{k_{\perp}^2 + \frac{\omega^2}{v^2} (1 - \beta^2 \epsilon(\omega))} \\ &= \frac{ze}{v\epsilon(\omega)} \sqrt{\frac{2}{\pi}} e^{i\omega z/v} \int_0^{+\infty} \frac{k_{\perp} J_0(k_{\perp} \rho)}{k_{\perp}^2 + \frac{\omega^2}{v^2} (1 - \beta^2 \epsilon(\omega))} dk_{\perp} \\ &= \frac{ze}{v\epsilon(\omega)} \sqrt{\frac{2}{\pi}} K_0 \left( \frac{|\omega| \rho}{v} \sqrt{1 - \beta^2 \epsilon(\omega)} \right) e^{i\omega z/v}, \end{aligned}$$

where we have used the following two integration identities,

$$J_0(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{ix \cos \phi} d\phi, \quad \int_0^{+\infty} \frac{x J_0(\alpha x)}{x^2 + \beta^2} dx = K_0(\alpha\beta), \quad \text{for } \beta > 0.$$

(b) If  $\epsilon(\omega)$  is a constant and  $1 - \beta^2 \epsilon > 0$ , we have

$$\begin{aligned} \Phi(\mathbf{x}, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \Phi(\omega, \mathbf{x}) e^{-i\omega t} d\omega \\ &= \frac{ze}{v\epsilon\pi} \int_{-\infty}^{+\infty} K_0 \left( \frac{|\omega| \rho}{v} \sqrt{1 - \beta^2 \epsilon} \right) e^{i\omega(z/v - t)} d\omega \\ &= \frac{2ze}{v\epsilon\pi} \int_0^{+\infty} K_0 \left( \frac{\omega \rho}{v} \sqrt{1 - \beta^2 \epsilon} \right) \cos \left( \omega \left( \frac{z}{v} - t \right) \right) d\omega \\ &= \frac{2ze}{v\epsilon\pi} \frac{\pi}{2} \left[ \frac{\rho^2}{v^2} (1 - \beta^2 \epsilon) + \left( \frac{z}{v} - t \right)^2 \right]^{-1/2} \\ &= \frac{ze/\epsilon}{\sqrt{(z - vt)^2 + \rho^2 (1 - \beta^2 \epsilon)}}, \end{aligned}$$

where we have used the identity

$$\int_0^{+\infty} K_0(\alpha x) \cos(\beta x) dx = \frac{\pi}{2\sqrt{\alpha^2 + \beta^2}}.$$

Define

$$\Gamma = \frac{1}{\sqrt{1 - \beta^2 \epsilon}},$$

the scalar potential can be written as

$$\Phi(\mathbf{x}, t) = \frac{ze\Gamma}{\epsilon\sqrt{\rho^2 + \Gamma^2(z - vt)^2}}.$$

This result is almost identical to Eq. (11.152), except the inclusion of the dielectric constant, which redefines  $\gamma$ . This result also reduces to Eq. (11.152) as  $\epsilon \rightarrow 1$ .

(c) When  $\beta^2 \epsilon > 1$ , we can still express the result formally as in part (a) as

$$\Phi(\omega, \mathbf{x}) = \frac{ze}{v\epsilon} \sqrt{\frac{2}{\pi}} K_0 \left( \frac{|\omega|\rho}{v} \sqrt{1 - \beta^2 \epsilon} \right) e^{i\omega z/v},$$

where the square root of  $1 - \beta^2 \epsilon < 0$  needs to be taken care of. By the general consideration of Section 7.10, namely, the imaginary part of the dielectric function  $\epsilon(\omega)$  is positive for  $\omega > 0$  and negative for  $\omega < 0$ , we can write

$$\sqrt{1 - \beta^2 \epsilon - i\omega 0^+} = \sqrt{\beta^2 \epsilon - 1} \exp \left( -\text{sgn}(\omega) \frac{\pi i}{2} \right),$$

then

$$\Phi(\omega, \mathbf{x}) = \frac{ze}{v\epsilon} \sqrt{\frac{2}{\pi}} K_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \cdot \exp \left( -\text{sgn}(\omega) \frac{\pi i}{2} \right) \right) e^{i\omega z/v}.$$

For  $\omega > 0$ , consider

$$K_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \cdot e^{-\pi i/2} \right).$$

Using Gradshteyn and Ryzhik, 8<sup>th</sup> ed., 8.406.1 and 8.405.1, pg. 920, we have

$$\begin{aligned} K_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \cdot e^{-\pi i/2} \right) &= \frac{\pi i}{2} H_0^{(1)} \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) \\ &= \frac{\pi i}{2} \left[ J_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) + iN_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) \right] \\ &= \frac{\pi}{2} \left[ -N_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) + iJ_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) \right]. \end{aligned}$$

Similarly, for  $\omega < 0$ , we can use 8.406.2 and 8.405.2 to process  $K_0$ . Finally, we have

$$\Phi(\omega, \mathbf{x}) = \frac{ze}{v\epsilon} \sqrt{\frac{\pi}{2}} \left[ -N_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) \pm iJ_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) \right], \quad (1)$$

for  $\omega \gtrless 0$ .

Perform the Fourier transform as in part (b),

$$\begin{aligned} \Phi(\mathbf{x}, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \Phi(\omega, \mathbf{x}) e^{-i\omega t} d\omega \\ &= \frac{ze}{2v\epsilon} \left\{ \int_0^{+\infty} \left[ -N_0 \left( \frac{\omega\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) + iJ_0 \left( \frac{\omega\rho}{v} \sqrt{\beta^2 \epsilon - 1} \right) \right] e^{i\omega(z/v - t)} d\omega \right. \end{aligned}$$

$$\begin{aligned}
& + \int_{-\infty}^0 \left[ -N_0 \left( -\frac{\omega\rho}{v} \sqrt{\beta^2\epsilon - 1} \right) - iJ_0 \left( -\frac{\omega\rho}{v} \sqrt{\beta^2\epsilon - 1} \right) \right] e^{i\omega(z/v-t)} d\omega \Big\} \\
& = \frac{ze}{v\epsilon} \int_0^{+\infty} \left[ -N_0 \left( \frac{\omega\rho}{v} \sqrt{\beta^2\epsilon - 1} \right) \cos \left( \omega \left( \frac{z}{v} - t \right) \right) \right. \\
& \quad \left. - J_0 \left( \frac{\omega\rho}{v} \sqrt{\beta^2\epsilon - 1} \right) \sin \left( \omega \left( \frac{z}{v} - t \right) \right) \right] d\omega.
\end{aligned}$$

We want to use the following two integration identities to evaluate the above integral,

$$\int_0^{+\infty} J_0(ax) \sin(bx) dx = \begin{cases} 0, & 0 < b < a \\ \frac{1}{\sqrt{b^2 - a^2}}, & 0 < a < b \end{cases}, \quad \int_0^{+\infty} N_0(ax) \sin(bx) dx = \begin{cases} 0, & 0 < b < a \\ -\frac{1}{\sqrt{b^2 - a^2}}, & 0 < a < b \end{cases}$$

From these we can see, if  $z > vt$ ,  $\Phi(\mathbf{x}, t) \equiv 0$ . This is consistent with the physical picture of Cherenkov radiation, that the potential vanishes for points ahead of the particle. On the other hand, if  $z < vt$ , we will have

$$\Phi(\mathbf{x}, t) = \frac{2ze}{v\epsilon} \left[ \left( \frac{z}{v} - t \right)^2 - \frac{\rho^2}{v^2} (\beta^2\epsilon - 1) \right]^{-1/2} = \frac{2ze/\epsilon}{\sqrt{(z - vt)^2 - (\beta^2\epsilon - 1)\rho^2}},$$

for  $|z - vt| > \rho\sqrt{\beta^2\epsilon - 1}$ , and 0 otherwise. This is constant with the result of vector potential in Section 13.4, which is related to the scalar potential by Eq. (13.25) for constant dielectric function,  $\mathbf{A}(\mathbf{x}, t) = \epsilon\beta\Phi(\mathbf{x}, t)$ .

In the above have used quite hand-waving argument to obtain Eq. (1). Instead, we can consider the familiar identity to write, for  $\beta^2\epsilon > 1$ ,

$$\begin{aligned}
\frac{1}{k_{\perp}^2 + \frac{\omega^2}{v^2}(1 - \beta^2\epsilon(\omega))} & \rightarrow \frac{1}{k_{\perp}^2 - \frac{\omega^2}{v^2}(\beta^2\epsilon(\omega) - 1) - i\text{sgn}(\omega)0^+} \\
& = \frac{1}{k_{\perp}^2 - \frac{\omega^2}{v^2}(\beta^2\epsilon(\omega) - 1)} + i\text{sgn}(\omega)\delta \left( k_{\perp}^2 - \frac{\omega^2}{v^2}(\beta^2\epsilon(\omega) - 1) \right) \\
& = \frac{1}{k_{\perp}^2 - \frac{\omega^2}{v^2}(\beta^2\epsilon(\omega) - 1)} + \frac{i}{2k_{\perp}} \text{sgn}(\omega)\delta \left( k_{\perp} - \frac{|\omega|}{v} \sqrt{\beta^2\epsilon(\omega) - 1} \right).
\end{aligned}$$

Upon integration, applying the identity

$$\int_0^{+\infty} \frac{xJ_0(ax)}{x^2 - b^2} dx = -\frac{\pi}{2} N_0(ab),$$

for  $a, b > 0$ , together with the delta function integration, we will have, for  $\beta^2\epsilon > 1$ ,

$$\int_0^{+\infty} \frac{k_{\perp} J_0(k_{\perp}\rho)}{k_{\perp}^2 + \frac{\omega^2}{v^2}(1 - \beta^2\epsilon(\omega))} dk_{\perp} = -\frac{\pi}{2} N_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2\epsilon - 1} \right) + \frac{i\pi}{2} \text{sgn}(\omega) J_0 \left( \frac{|\omega|\rho}{v} \sqrt{\beta^2\epsilon - 1} \right).$$

This is identical to the earlier result.